FloodVoice

Speak to be rescued. A voice-first flood reporting and rescue map for Chennai, with offline fallbacks.

Talk to Build AI Buildathon, 24 Oct 2026 · v2

1. The problem (researched)

Event	Impact	Source
2015 South India floods	500+ killed, about 1.8 million displaced	Wikipedia, "2015 South India floods"
Chennai 2015 (NIDM study)	More than 4 million people affected; loss of about USD 3 billion	NIDM report, 2021
Cyclone Michaung, Dec 2023	At least 13 killed in floods; 390,000+ people affected; power cut in flooded areas	Reuters, Wikipedia

> **The gap:** Tamil Nadu already has TN-SMART / TN-Alert, which pushes forecasts and alerts *to* people. There is no simple way for stranded people to send their exact location and need *back* to rescuers, especially in Tamil, by voice, or when the network is weak.

2. The user (Jobs-to-be-Done)

Lakshmi, 45, Velachery, 11 PM. Knee-deep water inside, an elderly mother who can't walk, the helpline is busy, phone at 20%.
Job: "Tell someone exactly where we are and what we need." **Feels:** helpless and unheard.
Second user: a rescue volunteer who needs to know where to go first.

3. Target

> How might we help stranded residents report flooding **by voice in about 10 seconds, in their own language**, so rescuers see the exact location and urgency **in under a minute**, even when the network is weak?

4. The solution: four layers

Layer	When	How it works	Today's demo
Online	Internet works	Speak in Tamil or English. AI (Gemini) extracts area, depth, need and severity. Pin appears on the live map.	Live
SMS	Weak signal, no data	4 big buttons (Rescue, Food, Medical, Road blocked) + phone GPS. One tap opens a pre-filled SMS such as `FV RESCUE 12.98,80.22 4ppl`.	Live (SMS app opens)
Mesh	No signal at all	Every phone has a FloodVoice ID. The report hops phone to phone over Bluetooth until one reaches signal.	Simulated (prototype)

Last resort	No neighbour nearby, phone dying	**Store & forward:** report waits on the phone and auto-sends when any signal or phone appears. **Went dark:** last known location saved while online; anyone silent 2+ hrs in a flood zone shows as "last seen here". **SOS beacon:** torch strobe + siren. **One tap 112.**	"Went dark" list live with seeded people; rest prototype

Why buttons offline: browser voice recognition in Chrome uses a server (MDN), so voice needs internet. Buttons + GPS work without it.
Proof the mesh idea is real: Bitchat, a Bluetooth mesh chat app, had 48,781 downloads in Nepal on 8 Sep 2025 during protests and social media bans (Yahoo Finance / Forbes).

5. AI features (Gemini)

Feature	What it does	Build time
AI Understands	Tanglish/Tamil/English report to JSON: area, depth (cm), category, severity 1-5, vulnerable yes/no, one-line reason	~15 min
AI Command Brief (wow)	One button: reads all open reports and writes a 3-line situation summary with where to send boats and medical teams first	~15 min
AI Triage	Rescue list sorted by priority with the "why" for each	~10 min

Priority score: `severity + depth/20 + 3 if vulnerable` (our own weighting).
Categories: Rescue, Medical, Food/Water, Electrical, Civic, with filter buttons per responder team.
Gemini prompt: *"You are a Chennai flood rescue triage AI. From this report, return only JSON with area, depth in centimetres, category (rescue, medical, food_water, electrical, civic), severity 1-5, vulnerable true/false, and reason in one short sentence."*

6. Features

- **Voice report** (Tamil / English / Tanglish) with a big mic button
- **AI extraction** into structured data: area, depth (ankle/knee/waist/chest), need, severity 1 to 5
- **Live rescue map** (Leaflet + OpenStreetMap) with pins coloured by severity
- **Volunteer dashboard**: requests sorted by urgency, "mark helping / rescued"
- **Trust signal**: nearby reports (within 1 km) merge into one pin with a count
- **Went dark list**: people who went silent in flood zones, with last known location
- **Offline mode**: 4 buttons + GPS + SMS fallback; status badge Online / SMS / Mesh
- **Bonus**: "stagnant water" reports become mosquito-risk zones after the flood

7. Architecture

Part	Tech
Frontend	React (Vite), Leaflet map, browser SpeechRecognition (lang ta-IN / en-IN), Geolocation API, `sms:` link
Backend	Node + Express, REST API
Database	MongoDB Atlas, collection `reports`
AI	Gemini: text to JSON {area, depth, need, severity}
Geocoding	OpenStreetMap Nominatim (or the phone's GPS)

Deploy	Vercel (frontend) + Render (backend)

Data model: `{ text, area, lat, lng, depth, need, severity, status, reportCount, category, vulnerable, reason, priority, channel: online|sms|mesh, lastSeenAt, createdAt }`
API: `POST /api/report` (text to AI to save) · `GET /api/reports` · `PATCH /api/report/:id` (status)

8. Who builds what (1 hour)

Tool	Task
Arena AI	Frontend UI: map, dashboard, mic, offline buttons, mesh hop animation
Cursor	Backend: Express + Mongo + Gemini, wire the frontend in
ChatGPT	15 seed reports as JSON (real Chennai areas, Tamil + English)
Tulaassi	Debugging, pitch, judges' Q&A

Order: 1. Map + 15 seed pins → 2. AI Understands (voice → Gemini → pin) → 3. AI Command Brief → 4. AI Triage list → 5. Offline buttons + SMS → 6. Went-dark list + mesh animation → 7. Backup video + rehearse.

9. Risks (premortem)

Risk	Fix
Venue Wi-Fi fails during the demo	Seed data loaded + recorded backup video; phone hotspot
Tamil voice recognition is poor	Test early; fall back to English or Tanglish in the demo
Judges think it's "just a map"	Open with Lakshmi's story; show offline layers
"Does Bluetooth really work?"	Say upfront: simulated today; real version uses Android nearby-device Bluetooth
Fake reports	Report count per spot; volunteers verify
AI misreads a report	Show the original text beside the AI result; volunteers can edit

10. Impact and next steps

- Rescuers reach the worst spots first; fewer missed calls for help.
- Next: native Android app with real Bluetooth mesh; SMS gateway number; partner with the city's disaster control room; Tamil voice prompts.

Sources: Wikipedia (2015 South India floods; Cyclone Michaung), NIDM "Chennai Flood 2015" (2021), Reuters (5 Dec 2023), The Hindu and app stores (TN-SMART / TN-Alert), MDN (SpeechRecognition), Yahoo Finance and Forbes (Bitchat, Sep 2025). Figures are from search results read on 24 Oct 2026; death counts vary by source.